Energy-Norm Error Bound Proposition

Small split energy controls the final energy error

u_ϕ

$u_{\text{pred}} = \frac{1}{2}(u_\phi + u_\psi)$

u_ψ

DIFFUSION-WEIGHTED ENERGY NORM

$$\|v\|_a^2 := \int_\Omega a(x)|\nabla v(x)|^2\,dx, \qquad a(x) > 0.$$

Here $a(x)$ is the diffusion coefficient.

SPLIT-ENERGY LOSS

$$\mathcal{E}_{\text{split}} = \|u_\phi - u_\psi\|_a^2.$$

REFERENCE SOLUTION

$$\mathcal{L}u_* = f, \qquad u_*|_{\partial\Omega} = 0.$$

$$\|u_{\text{pred}} - u_*\|_a \le \frac{C_E}{2}\sqrt{\mathcal{E}_{\text{split}}}.$$

C_E: stability constant of the fixed elliptic operator.

Conditional on exact/controlled Green reconstruction and $H_0^1(\Omega)$-admissible represented solutions.